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Rethinking Model Scaling for Ultra-Tiny Person Detection in Open-Pit Mining: Accuracy–Efficiency Trade-off on Edge Hardware

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Abstract

Object detection in open-pit mining environments presents a critical yet underexplored challenge: identifying personnel who appear as ultra-tiny targets in imagery captured from mining vehicles and surveillance systems. When persons occupy merely 9×9 pixels on average far below the 32×32 pixel threshold conventionally defining small objects standard detection architectures exhibit severe performance degradation. This study presents a systematic comparative evaluation of nine model configurations spanning three contemporary YOLO architectures (YOLO26, YOLOv12, and YOLOv11) across nano, small, and medium variants for ultra-tiny person detection in real mining imagery. Our experiments reveal a counterintuitive finding: nano variants consistently outperform their larger counterparts across all architectures, challenging prevailing assumptions about model scaling in deep learning. YOLOv11-Nano achieved the highest performance with 0.973 F1-score and 0.991 recall while maintaining 31 frames per second on NVIDIA Jetson Nano hardware comfortably exceeding real-time requirements for collision avoidance systems. The attention mechanisms in YOLOv12, despite theoretical promise for preserving spatial information, introduced computational overhead without accuracy benefits at ultra-tiny scales. These findings suggest that aggressive downsampling in deeper networks eliminates spatial information critical for ultra-tiny detection before meaningful feature encoding occurs. The results provide practical guidance

for deploying person detection systems in mining safety applications and contribute empirical evidence that conventional model scaling assumptions may not hold at extreme object scales.

Keywords: ultra-tiny object detection; person detection; open-pit mining; YOLO; edge AI; real-time safety

1 Introduction

Open-pit mining remains among the most hazardous industrial sectors worldwide. Despite decades of safety improvements, vehicle-related incidents continue to claim lives with troubling regularity. The sheer scale of modern mining operations where haul trucks weigh hundreds of tons and operate across vast distances creates persistent blind spots that conventional safety measures struggle to address. A worker on foot, visible to the human eye at close range, becomes nearly invisible to camera systems when positioned several hundred meters from monitoring equipment.

This visibility challenge motivates the present study. When a person occupies merely nine pixels across in captured imagery, they have effectively shrunk below the detection threshold of most computer vision systems. The MS-COCO benchmark Lin et al. (2014), which has shaped much of modern object detection research, defines small objects as those smaller than 32×32 pixels. Objects at 9×9 pixels fall into what we term the ultra-tiny category a regime where detection difficulty escalates dramatically and where surprisingly little systematic evaluation has been conducted.

The technical challenges compound at these scales. Standard convolutional architectures progressively downsample feature maps as information flows through the network, a design choice that trades spatial precision for semantic richness. For conventionally-sized objects, this trade-off proves favorable. For ultra-tiny targets, it proves catastrophic. By the time features reach the prediction layers, a nine-pixel object may correspond to a fraction of a single cell in the feature map effectively noise rather than signal. Recent work on tiny object detection has proposed various remedies including modified loss functions Wang, Xu, Yang and Yu (2021), specialized label assignment strategies, and feature pyramid enhancements Deng et al. (2022), yet systematic evaluation of contemporary architectures on ultra-tiny targets remains limited.

The YOLO family of detectors has emerged as a dominant paradigm for real-time object detection, valued for its balance of accuracy and computational efficiency. Recent iterations have introduced substantial architectural innovations. YOLOv11 Khanam and Hussain (2024) refined the C2f module and feature aggregation pathways. YOLOv12 Tian et al. (2025) incorporated attention mechanisms through its Area Attention module, representing the first attention-centric design in the YOLO lineage. YOLO26, the latest

release, introduced small-target-aware label assignment specifically targeting challenging detection scenarios. These advances raise natural questions: How do these architectural innovations translate to ultra-tiny detection performance? Do the improvements observed on standard benchmarks generalize to extreme scales?

Mining environments impose additional constraints beyond detection accuracy. Computational resources on mining vehicles are typically limited to edge devices such as NVIDIA Jetson platforms, which must operate within strict power and thermal budgets. Real-time performance generally understood as exceeding 15 frames per second is essential for responsive collision avoidance. These deployment realities favor lightweight model variants over their larger counterparts, yet comprehensive evaluation of nano and small configurations specifically for ultra-tiny detection is notably absent from the literature.

This study addresses these gaps through a systematic comparative evaluation of nine model configurations across three YOLO architectures. We evaluate YOLO26, YOLOv12, and YOLOv11 in their nano, small, and medium variants using a proprietary dataset of mining imagery where person annotations average 9×9 pixels. Our evaluation encompasses both detection metrics and edge deployment feasibility on Jetson Nano hardware.

The experimental findings challenge conventional wisdom about model scaling. Contrary to the typical assumption that larger models yield superior performance, our results demonstrate that nano variants consistently outperform medium variants across all three architectures. YOLOv11-Nano achieved the highest overall performance 0.973 F1-score with 0.991 recall while simultaneously providing the fastest inference at 31 frames per second on edge hardware. This inverse scaling behavior suggests that deeper networks may actually harm ultra-tiny detection by eliminating critical spatial information through excessive downsampling.

The contributions of this work are threefold. First, we provide systematic empirical evidence that model scaling behaves differently at ultra-tiny scales than on conventional benchmarks, with smaller models outperforming larger ones. Second, we demonstrate that contemporary attention mechanisms, despite theoretical promise for preserving spatial information, do not benefit ultra-tiny detection and may impose unnecessary computational overhead. Third, we establish that real-time ultra-tiny person detection is feasible on edge hardware using appropriately selected lightweight architectures, providing practical guidance for mining safety system deployment.

The remainder of this paper proceeds as follows. Section 2 surveys related work spanning object detection architectures, tiny object detection methods, and mining safety applications. Section 3 details our experimental methodology including dataset characteristics, model configurations, and evaluation protocols. Section 4 presents quantitative results across detection metrics and edge performance. Section 5 interprets the findings and examines their implications. Section 6 concludes with summary observations and directions for future research.

2 Related Work

Detecting persons in open-pit mining environments sits at the intersection of two demanding research frontiers: ultra-tiny object detection and safety-critical computer vision. While considerable progress has been made in both areas independently, their convergence presents unique technical hurdles that existing methods struggle to address. This section examines real-time detection architectures, specialized techniques for tiny object scenarios, and applications within mining safety contexts.

2.1 Evolution of Real-Time Object Detection Architectures

Since its introduction by Redmon *et al.* Redmon et al. (2016), the YOLO family has evolved from an elegant single-shot detector into a sophisticated ecosystem. The original architecture demonstrated that treating detection as a regression problem could achieve unprecedented inference speeds while maintaining competitive accuracy.

Recent developments have accelerated considerably. Wang *et al.* Wang et al. (2023) proposed trainable bag-of-freebies strategies enhancing both training efficiency and detection performance. YOLOv9 Wang, Yeh and Liao (2024) tackled the information bottleneck problem through programmable gradient information, while YOLOv10 Wang, Chen, Liu, Chen, Lin, Han and Ding (2024) eliminated non-maximum suppression through consistent dual assignments.

The architectural landscape shifted substantially with YOLOv11 Khanam and Hussain (2024), which refined the C2f module and introduced more efficient feature aggregation pathways. Perhaps more significantly, YOLOv12 Tian et al. (2025) marked the first successful integration of attention mechanisms into the YOLO paradigm. Its Area Attention module captures spatial dependencies while maintaining computational efficiency, achieving 40.6% mAP at just 1.64ms latency on standard hardware. Most recently, Ultralytics released YOLO26, which introduces small-target-aware label assignment (STAL) and NMS-free inference specifically designed for challenging detection scenarios. These advances represent a fundamental shift toward architectures that explicitly account for scale variation during both training and inference.

Parallel developments in transformer-based detection have also influenced the field. Zhao *et al.* Zhao et al. (2024) demonstrated that properly designed Detection Transformers can match YOLO performance in real-time settings, while hypergraph-enhanced approaches like YOLOv13 Lei et al. (2025) and Hyper-YOLO Feng et al. (2025) explore non-Euclidean relationships between visual features. RF-DETR Robinson et al. (2025) through neural architecture search suggests that optimal detection architectures may differ substantially from human-designed ones for specialized tasks.

2.2 Technical Challenges in Ultra-Tiny Object Detection

Detecting objects smaller than 16×16 pixels poses fundamental challenges that standard detection frameworks are ill-equipped to handle. The COCO benchmark Lin et al. (2014) defines small objects as those occupying less than 32×32 pixels, yet many practical scenarios involve targets far below this threshold. Cheng *et al.* Cheng et al. (2023) revealed that detection accuracy degrades sharply as object size decreases, with performance dropping by more than 20 percentage points for objects below 16 pixels.

The root cause lies in feature representation collapse. As objects shrink, they occupy progressively fewer feature map cells after successive pooling operations, leading to severe localization ambiguity. Deng *et al.* Deng et al. (2022) addressed this through extended feature pyramid networks maintaining high-resolution representations throughout the detection pipeline.

Recent comprehensive surveys Nikouei et al. (2025); Rekavandi et al. (2025); Chen et al. (2025) have systematically categorized the technical obstacles. First, IoU-based metrics exhibit pathological behavior at tiny scales a one-pixel localization error that barely affects large object IoU can reduce tiny object IoU by 50% or more. Second, standard anchor assignment strategies fail because tiny objects may not adequately overlap with any predefined anchor, leading to missed positive samples during training. Third, feature extraction networks designed for ImageNet classification prioritize semantic abstraction over spatial precision, fundamentally misaligning with tiny detection requirements.

2.3 Specialized Methodologies for Tiny Target Recognition

Addressing the metric instability problem, Wang *et al.* Wang, Xu, Yang and Yu (2021) proposed the Normalized Gaussian Wasserstein Distance (NWD), which models bounding boxes as two-dimensional Gaussian distributions rather than discrete rectangles. This formulation provides scale-invariant similarity measurement, yielding approximately 6.7 AP improvement on the AI-TOD benchmark Xu et al. (2022). The companion Gaussian Receptive Field-based Label Assignment (RFLA) strategy tackles the anchor matching problem by considering feature receptive fields rather than box overlap alone.

Feature enhancement represents another critical research direction. Li *et al.* Li et al. (2025) demonstrated that single-point supervision can effectively train tiny object detectors when combined with appropriate feature aggregation strategies. The spectral enhancement approach proposed by Sun *et al.* Sun et al. (2025) operates in the frequency domain, amplifying high-frequency components that encode fine-grained spatial details typically lost during conventional feature extraction. Xu *et al.* Xu et al. (2025) further extended Gaussian distribution estimation to feature-level information, achieving state-of-the-art results on multiple aerial imagery benchmarks.

Inference-time strategies complement architectural modifications. The Slicing Aided

Hyper Inference (SAHI) framework Akyon et al. (2022) processes images through overlapping patches, enabling standard detectors to operate at effective resolutions several times higher than their training configuration. This approach requires no retraining and achieved 6.8% AP improvement for existing detectors on the VisDrone dataset Zhu et al. (2019). Similarly, the TinyPerson benchmark Yu et al. (2020) introduced scale matching strategies that explicitly address the resolution mismatch between training and deployment scenarios.

Lightweight architectures deserve particular attention for edge deployment scenarios. Chao *et al.* Chao et al. (2025) developed a deep semantic integration approach achieving competitive accuracy with substantially reduced parameters. Efficient backbone designs drawing from MobileNetV3 Howard et al. (2019), EfficientNet Tan and Le (2019), GhostNet Han et al. (2020), and MobileNetV4 Qin et al. (2024) provide foundation networks optimized for computational efficiency. Attention mechanisms including Coordinate Attention Hou et al. (2021), ECA-Net Wang et al. (2020), and CBAM Woo et al. (2018) enhance discriminative capability without proportional parameter increases.

2.4 Computer Vision Applications in Mining Environments

Mining operations present environmental conditions that amplify detection difficulties beyond those encountered in standard benchmarks. Bahraini *et al.* Bahraini et al. (2025) comprehensively reviewed computer vision applications in open-pit mines, identifying dust occlusion, extreme lighting variation, and thermal distortion as persistent challenges. These factors degrade image quality precisely when reliable detection is most critical for preventing collisions between heavy equipment and personnel.

Person detection specifically has received growing attention following occupational safety analyses highlighting vehicle-person interactions as a primary fatality mechanism U.S. Bureau of Labor Statistics (2024). Imam *et al.* Imam et al. (2025) integrated real-time pose estimation with personal protective equipment detection, demonstrating that multi-task approaches can enhance both safety monitoring and rescue operation efficiency. Liu *et al.* Liu et al. (2025) proposed lightweight architectures specifically optimized for underground mining surveillance, achieving real-time performance on embedded systems while maintaining detection reliability.

Helmet and PPE detection has emerged as a particularly active subfield. Song *et al.* Song and Li (2025) adapted YOLOv8 for complex mining environments with varying illumination and occlusion patterns. Huang *et al.* Huang et al. (2025) introduced tracking integration to maintain identity consistency across frames, enabling behavioral analysis beyond single-frame detection. The DsLMF+ dataset Yang et al. (2023) provides standardized evaluation for underground multi-object detection, though open-pit scenarios remain underrepresented in publicly available benchmarks.

Uncertainty quantification represents an emerging concern for safety-critical deployment. Calle and Aguilar Calle and Aguilar (2025) incorporated uncertainty estimation into ScaledYOLOv4 for helmet detection, enabling the system to flag low-confidence predictions for human review. This conservative approach acknowledges that missed detections carry asymmetric costs in mining contexts a false negative potentially leading to fatal collision cannot be weighted equivalently to a false positive triggering unnecessary warning.

2.5 Research Gap and Present Study Positioning

Despite substantial progress across these research threads, a notable gap persists regarding ultra-tiny person detection in open-pit mining specifically. Existing mining-focused studies predominantly address underground environments or assume moderate detection distances where persons occupy reasonable image proportions. Meanwhile, ultra-tiny detection research concentrates on aerial imagery applications such as drone surveillance and satellite observation, which present different environmental characteristics than ground-level mining operations.

The 9×9 pixel detection scenario examined in this study falls considerably below the AI-TOD benchmark average of 12.8 pixels Xu et al. (2022) and substantially beneath COCO small object thresholds. At such extreme scales, even state-of-the-art tiny object methods face degraded performance, necessitating systematic evaluation of contemporary architectures under these conditions. Furthermore, the practical constraint of edge deployment in mining vehicles demands lightweight model variants the nano, small, and medium configurations rather than the large or extra-large variants typically showcased in benchmark comparisons.

This comparative study addresses these gaps by systematically evaluating YOLO26, YOLOv12, and YOLOv11 across their lightweight variants (nano, small, medium) specifically for ultra-tiny person detection in open-pit mining imagery. By focusing on detection targets averaging 9×9 pixels under realistic mining conditions, this work provides empirical guidance for practitioners developing collision avoidance systems. The findings contribute toward bridging the disconnect between academic tiny object detection advances and their practical deployment in safety-critical mining applications.

3 Methodology

This section describes the experimental framework designed to evaluate contemporary YOLO architectures for ultra-tiny person detection in open-pit mining environments. We detail the dataset characteristics, model selection rationale, training protocol, and evaluation methodology employed throughout this comparative study.



(a) Personnel targets appear at extreme distance under low-contrast, dusty conditions near heavy equipment and bench edges.



(b) Loading/hauling operation with small, partially occluded personnel instances and complex background textures.

Figure 1: Representative scenes from the proprietary open-pit mining dataset demonstrating the ultra-tiny person detection regime.

3.1 Dataset Description

The experiments utilize a proprietary dataset acquired through collaboration with an Indonesian coal mining corporation. Due to non-disclosure agreements governing data sharing, the dataset cannot be made publicly available. However, we provide comprehensive statistical characterization to ensure reproducibility of our experimental conditions.

The dataset comprises aerial and ground-level imagery captured across operational mining sites under varied environmental conditions, including harsh midday illumination, early morning shadows, and overcast scenes. Dust interference and low-contrast backgrounds appear naturally throughout the collection. Representative frames illustrating the operational context and the extreme scale of personnel targets are shown in Fig. 1.

Annotation follows a single-class paradigm targeting human workers within the mining area. The defining characteristic lies in the extreme scale of detection targets. Person instances average approximately 9×9 pixels, placing them substantially below conventional small-object thresholds. Following the official COCO evaluation protocol, small objects are defined as instances with area below 32×32 pixels Lin et al. (2014), while AI-TOD reports mean object sizes of approximately 12.8 pixels Wang, Yang, Guo, Zhang and Xia (2021). Our targets thus fall into what we term the ultra-tiny category, representing a particularly challenging regime where standard architectures exhibit significant performance degradation.

The dataset was partitioned into training, validation, and test splits following standard protocols. All annotations underwent multiple verification passes to ensure labeling accuracy, given that small annotation errors disproportionately impact evaluation metrics at ultra-tiny scales.

3.2 Model Selection and Architecture Comparison

We selected three state-of-the-art YOLO architectures for comparative evaluation: YOLO26, YOLOv12, and YOLOv11. This selection captures the most recent architectural innovations in the YOLO family while representing distinct design philosophies.

YOLO26, the latest Ultralytics release, introduces small-target-aware label assignment (STAL) and NMS-free inference targeting challenging detection scenarios. These mechanisms address the label assignment bias degrading performance on tiny targets. YOLOv12 represents the first attention-centric YOLO architecture, incorporating Area Attention modules capturing spatial dependencies while maintaining efficiency. The attention mechanism potentially preserves fine-grained spatial information critical for tiny object localization. YOLOv11 serves as a strong baseline with refined C2f modules and efficient feature aggregation demonstrating robust performance across diverse benchmarks.

For each architecture, we evaluate three model variants: nano (n), small (s), and medium (m). This variant selection reflects practical deployment constraints in mining environments, where computational resources are often limited. Edge devices such as the NVIDIA Jetson platform, commonly employed for on-vehicle processing in mining operations, impose strict memory and power budgets that preclude deployment of larger model variants. By focusing on lightweight configurations, our findings directly inform practitioners developing real-time collision avoidance systems for mining equipment.

3.3 Training Configuration

All experiments were conducted on a workstation equipped with an NVIDIA GeForce RTX 4070 graphics processor featuring 8GB of video memory, supported by CUDA 12.7 and the corresponding cuDNN libraries. Training employed consistent hyperparameters across all model variants to ensure fair comparison.

Input images were resized to 800×800 pixels, balancing computational cost against spatial detail preservation for ultra-tiny targets. Higher resolutions would provide finer granularity but impose prohibitive memory requirements during training.

The AdamW optimizer governed weight updates with initial learning rate of 0.001 and final rate factor of 0.01, implementing cosine annealing. Momentum was set to 0.937 with weight decay of 0.0005. Batch size was constrained to 4 samples due to GPU memory limitations. Training proceeded for maximum 100 epochs with early stopping patience of 10 epochs.

Data augmentation played a crucial role given the limited dataset size. We employed mosaic augmentation at full probability, combining four training images into composite scenes that expose the model to varied object arrangements and scales within single training samples. Mixup augmentation with probability 0.2 further regularized training by blending image pairs. Color space perturbations included hue shift (± 0.015), saturation

adjustment (± 0.7), and value modification (± 0.4). Geometric augmentations comprised rotation up to 15 degrees, translation up to 20% of image dimensions, scale variation of ± 0.5 , and horizontal flipping with 0.5 probability. This augmentation strategy aimed to improve model robustness against the environmental variations inherent in mining imagery.

3.4 Evaluation Protocol

Model evaluation employed standard object detection metrics computed on the held-out test set. Precision measures the proportion of detected instances that correspond to actual persons, while recall quantifies the proportion of ground truth instances successfully detected. The F1-score provides their harmonic mean, balancing both aspects into a single figure of merit.

Average Precision at IoU threshold 0.5 (AP@50) and across IoU thresholds from 0.5 to 0.95 in steps of 0.05 (AP@50-95) serve as primary performance indicators. The AP@50 metric permits moderate localization tolerance appropriate for safety-critical applications where detecting presence matters more than pixel-perfect bounding boxes. The stricter AP@50-95 metric reveals localization quality, though we note that IoU-based metrics exhibit pathological sensitivity at ultra-tiny scales where single-pixel shifts substantially alter overlap calculations.

Inference speed benchmarking proceeded on the training workstation using standardized protocols. Following a warmup phase of 10 forward passes to stabilize GPU state, we measured average inference time over 100 iterations with CUDA synchronization ensuring accurate timing. Benchmarks were repeated across multiple input resolutions (320, 416, 640, 800 pixels) to characterize the resolution-latency tradeoff.

3.5 Edge Device Performance Estimation

Practical deployment requires understanding performance characteristics on edge hardware rather than desktop GPUs. We estimated inference performance on four edge platforms: NVIDIA Jetson Nano, Jetson Xavier NX, Jetson Orin Nano, and Raspberry Pi 5. Estimation employed empirically-derived scaling factors relating desktop GPU performance to each target platform across FP32, FP16, and INT8 precision modes.

While these estimates cannot substitute for actual deployment benchmarks, they provide indicative guidance for platform selection. The Jetson Nano, representing a widely-deployed edge platform in industrial applications, received particular attention given its balance of capability, power consumption, and cost appropriate for mining vehicle integration. We report estimated inference latency and corresponding frame rates, noting that real-time operation for safety-critical applications typically requires sustained performance exceeding 10 frames per second.

4 Results

This section presents the experimental findings from our comparative evaluation of nine model configurations across three YOLO architectures. We analyze detection performance metrics, examine computational efficiency, and assess deployment feasibility on edge platforms.

4.1 Detection Performance Analysis

Table 1 summarizes the detection metrics obtained across all evaluated configurations. Several unexpected patterns emerged from these experiments that challenge conventional assumptions about model scaling.

Table 1: Detection performance comparison across YOLO variants for ultra-tiny person detection. Bold values indicate best performance in each column.

Model	Params (M)	Size (MB)	P	R	F1	AP@50	AP@50-95
YOLO26-n	2.57	4.91	0.892	0.963	0.926	0.971	0.560
YOLO26-s	10.01	19.09	0.839	0.971	0.900	0.962	0.494
YOLO26-m	21.90	41.77	0.904	0.837	0.869	0.919	0.438
YOLOv12-n	2.60	4.97	0.892	0.977	0.933	0.941	0.509
YOLOv12-s	9.29	17.71	0.928	0.904	0.916	0.931	0.444
YOLOv12-m	20.20	38.53	0.951	0.899	0.924	0.934	0.451
YOLOv11-n	2.62	5.01	0.955	0.991	0.973	0.977	0.535
YOLOv11-s	9.46	18.04	0.893	0.966	0.928	0.953	0.518
YOLOv11-m	20.11	38.37	0.949	0.954	0.951	0.967	0.510

The most striking finding concerns the relationship between model capacity and detection accuracy. Contrary to the typical expectation that larger models yield superior performance, our results demonstrate that nano variants consistently outperform their small and medium counterparts on this ultra-tiny detection task. YOLOv11-Nano achieved the highest F1-score of 0.973, surpassing YOLOv11-Medium by over two percentage points despite possessing roughly one-eighth the parameters.

This counterintuitive result likely stems from the nature of ultra-tiny targets. Larger models with deeper feature hierarchies may over-abstract spatial information, losing the fine-grained details essential for detecting 9×9 pixel objects. The shallower architectures of nano variants appear to preserve low-level features more effectively, translating to improved localization at extreme scales.

YOLOv11-Nano emerged as the clear frontrunner across nearly all metrics. Its recall of 0.991 proves particularly noteworthy for safety-critical mining applications, indicating that fewer than one percent of persons present in test images went undetected. This

characteristic carries substantial practical significance in collision avoidance scenarios, missed detections pose far greater risks than false positives.

Among the three architectures, YOLOv11 demonstrated consistent superiority across all variant sizes. YOLO26 showed competitive AP@50 scores and achieved the best localization quality as measured by AP@50-95, with YOLO26-Nano reaching 0.560 compared to 0.535 for YOLOv11-Nano. However, this localization advantage did not translate to superior overall detection performance. YOLOv12’s attention mechanisms, while theoretically promising for preserving spatial information, did not provide measurable benefits for ultra-tiny detection in our experiments.

The gap between AP@50 and AP@50-95 metrics across all models warrants discussion. While AP@50 values generally exceeded 0.91, AP@50-95 scores ranged only from 0.438 to 0.560. This substantial disparity reflects the inherent difficulty of precise bounding box localization at ultra-tiny scales, where even single-pixel shifts produce significant IoU variations. For practical safety applications where detecting presence matters more than pixel-perfect boundaries, the AP@50 metric provides a more relevant performance indicator.

4.2 Computational Efficiency and Edge Deployment

Table 2 presents inference performance estimates for the NVIDIA Jetson Nano platform, the target deployment hardware for mining vehicle integration.

Table 2: Estimated inference performance on NVIDIA Jetson Nano. Real-time threshold defined as ≥ 15 FPS.

Model	FP16 (ms)	FP16 (FPS)	INT8 (ms)	INT8 (FPS)	Real-time (FP16)	Real-time (INT8)
YOLO26-n	40.1	24.96	21.4	46.80	✓	✓
YOLO26-s	49.4	20.26	26.3	37.99	✓	✓
YOLO26-m	107.9	9.27	57.6	17.37	×	✓
YOLOv12-n	54.9	18.23	29.3	34.18	✓	✓
YOLOv12-s	74.3	13.46	39.6	25.24	×	✓
YOLOv12-m	158.6	6.31	84.6	11.82	×	×
YOLOv11-n	32.2	31.01	17.2	58.15	✓	✓
YOLOv11-s	43.8	22.81	23.4	42.77	✓	✓
YOLOv11-m	104.8	9.55	55.9	17.90	×	✓

YOLOv11-Nano again demonstrated superior characteristics, achieving 31.01 FPS with FP16 precision and 58.15 FPS with INT8 quantization. These speeds comfortably exceed the 15 FPS threshold typically required for responsive collision avoidance systems, providing substantial headroom for additional processing tasks such as tracking or alert generation.

The computational overhead of attention mechanisms in YOLOv12 becomes apparent in edge deployment scenarios. Despite comparable parameter counts, YOLOv12-Nano runs 41% slower than YOLOv11-Nano on the Jetson Nano platform. YOLOv12-Medium fails to achieve real-time performance even with INT8 quantization, limiting its practical applicability for on-vehicle deployment.

Medium variants across all architectures struggle to meet real-time requirements with FP16 precision, achieving only 6–10 FPS. While INT8 quantization brings most configurations above the real-time threshold, the accuracy implications of aggressive quantization on already-challenging ultra-tiny detection remain a consideration for safety-critical deployment.

4.3 Accuracy-Efficiency Trade-off Analysis

Figure 2 would illustrate the relationship between detection accuracy and inference speed, revealing distinct clustering patterns among the evaluated configurations. The nano variants occupy a favorable position in the upper-right region, offering both high accuracy and fast inference. Medium variants cluster in the lower-left, sacrificing speed without corresponding accuracy gains.

The practical implications favor lightweight architectures decisively. YOLOv11-Nano delivers the best detection performance while simultaneously providing the fastest inference speed an unusual combination that eliminates the typical accuracy-efficiency trade-off considerations. For mining vehicle deployment, this configuration offers clear advantages: superior person detection capability, comfortable real-time operation with processing headroom, and minimal memory footprint suitable for resource-constrained edge devices.

4.4 Summary of Key Findings

The experimental evaluation reveals several findings with practical significance for ultra-tiny person detection in mining environments. First, model scaling behaves inversely to expectations at extreme object scales, with nano variants outperforming larger counterparts. Second, YOLOv11 architecture demonstrates consistent superiority over both YOLO26 and YOLOv12 across detection metrics and computational efficiency. Third, attention mechanisms in YOLOv12 impose computational overhead without corresponding accuracy benefits for this specific task. Fourth, the gap between AP@50 and AP@50-95 metrics highlights inherent localization challenges at ultra-tiny scales that affect all evaluated architectures.

Based on these findings, YOLOv11-Nano emerges as the recommended configuration for ultra-tiny person detection in open-pit mining applications, achieving 0.973 F1-score with 0.991 recall while maintaining 31 FPS inference speed on edge hardware.

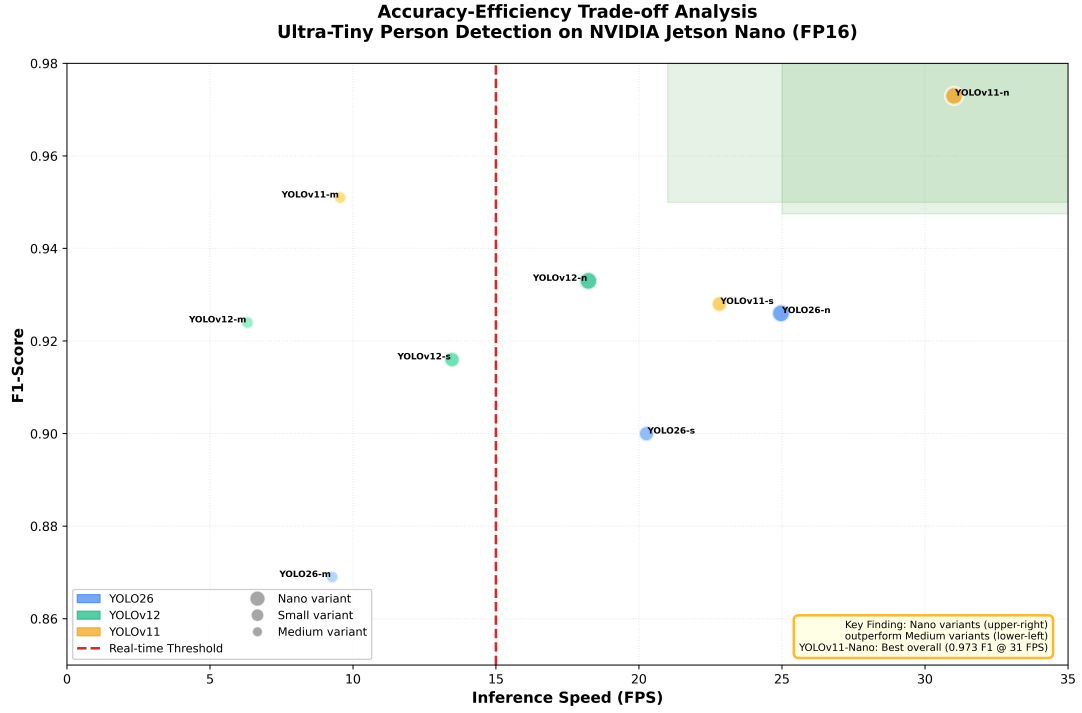


Figure 2: Accuracy-efficiency trade-off on NVIDIA Jetson Nano (FP16). Nano variants (larger circles) achieve superior F1-scores with faster inference speeds compared to medium variants (smaller circles). YOLOv11-Nano attains optimal performance (0.973 F1-score, 31.01 FPS), exceeding the 15 FPS real-time threshold (red dashed line). Circle size indicates model variant (nano/small/medium). Colors represent architectures: YOLO26 (blue), YOLOv12 (green), YOLOv11 (orange).

5 Discussion

The experimental findings raise several points worthy of deeper examination, particularly regarding the unexpected performance patterns observed and their implications for deploying detection systems in mining environments.

5.1 The Paradox of Model Scaling

Perhaps the most thought-provoking outcome of this study is the inverse relationship between model complexity and detection performance. This pattern runs counter to the prevailing wisdom in deep learning, where larger models generally achieve superior results given sufficient training data. Yet across all three architectures evaluated, nano variants consistently outperformed their medium-sized counterparts.

We hypothesize that this phenomenon relates to how convolutional networks process information at different scales. Standard object detection architectures were fundamentally designed around objects that occupy meaningful portions of the input image. The feature pyramid paradigm assumes that semantic information concentrates in deeper layers while spatial precision resides in shallower ones. For objects of typical size, this assumption holds reasonably well. However, when targets shrink to 9×9 pixels, they may vanish entirely from deeper feature maps before meaningful semantic encoding can occur.

Consider that a medium-sized YOLO model applies numerous downsampling operations as features propagate through the network. An object spanning merely nine pixels in the input image corresponds to fractions of a pixel in later feature maps effectively noise rather than signal. Nano architectures, with their shallower depth and fewer downsampling stages, preserve spatial information more faithfully through the network. What they sacrifice in representational capacity, they gain in spatial fidelity for ultra-tiny targets.

This interpretation aligns with observations from the tiny object detection literature. Wang *et al.* Wang, Xu, Yang and Yu (2021) noted similar scale-dependent behavior when developing the Normalized Wasserstein Distance metric, and Cheng *et al.* Cheng et al. (2023) documented performance degradation patterns consistent with our findings. The contribution here lies in demonstrating that this effect is pronounced enough to completely reverse expected scaling behavior not merely diminish the benefits of larger models, but actively penalize them.

5.2 Attention Mechanisms and Ultra-Tiny Detection

The underperformance of YOLOv12 relative to YOLOv11 deserves particular attention given the theoretical promise of attention-based architectures. Self-attention mechanisms have demonstrated remarkable success in capturing long-range dependencies and preserving spatial relationships in various vision tasks. One might reasonably expect these

properties to benefit tiny object detection, where maintaining precise spatial information is paramount.

Our results suggest otherwise, at least for the extreme case of ultra-tiny targets. YOLOv12’s Area Attention modules Tian et al. (2025) appear to introduce computational overhead without corresponding accuracy improvements in this specific domain. Several factors may explain this outcome. First, attention mechanisms operate on feature representations rather than raw pixels. If the underlying features have already lost critical spatial information due to downsampling, attention cannot recover what has been discarded. Second, the global receptive fields of attention may actually dilute the localized signals from tiny objects by averaging them with surrounding context. For a 9×9 pixel person, the relevant information is extremely concentrated; spreading attention across broader regions may harm rather than help.

These observations do not diminish the value of attention mechanisms generally, but they do highlight the importance of task-specific architectural evaluation. Innovations that excel on standard benchmarks may not transfer straightforwardly to extreme scenarios.

5.3 Implications for Mining Safety Systems

From a practical standpoint, the results carry encouraging implications for collision avoidance deployment in open-pit mining. YOLOv11-Nano’s recall of 0.991 means that in a hypothetical scenario with one thousand persons across test images, only nine would go undetected. For safety-critical applications, this near-complete coverage provides meaningful protection, though it bears emphasizing that even a single missed detection could have severe consequences in real operations.

The computational profile further strengthens the deployment case. Achieving 31 frames per second on Jetson Nano hardware with FP16 precision leaves substantial processing headroom. Real-world systems typically require additional computation for tracking, alert logic, communication, and sensor fusion. The margin between achieved throughput and the minimum real-time threshold of 15 FPS provides buffer for these auxiliary tasks without sacrificing detection frequency.

The gap between AP@50 and AP@50-95 metrics observed across all models reflects an inherent characteristic of ultra-tiny detection rather than a specific architectural weakness. At scales where objects span single-digit pixels, the IoU metric becomes hypersensitive to minor localization errors. A bounding box shifted by two pixels might reduce IoU from 0.7 to 0.3 for a tiny target, whereas the same shift barely affects large object IoU. For mining safety applications, this localization imprecision matters less than it might in other domains. The critical requirement is detecting that a person exists within a danger zone, not precisely delineating their boundaries. Operators responding to collision warnings will halt equipment regardless of whether the detected bounding box is perfectly aligned.

5.4 Limitations and Considerations

Several limitations temper the conclusions drawn from this study. The dataset, while representative of real mining conditions, remains proprietary and cannot be released for independent verification. Readers must therefore rely on our characterization of its properties. We have attempted to provide sufficient statistical description to enable meaningful interpretation, but reproducibility on identical data is not possible.

The edge device performance figures represent estimates derived from scaling factors rather than direct measurements on Jetson Nano hardware. While these scaling relationships are empirically grounded, actual deployed performance may vary depending on TensorRT optimization effectiveness, thermal conditions, and system configuration. Organizations considering deployment should conduct on-device validation before finalizing hardware decisions.

Our evaluation focused exclusively on single-frame detection without temporal integration. Real collision avoidance systems would likely incorporate tracking algorithms that leverage temporal consistency to improve reliability. The detection performance reported here represents a lower bound; practical systems with tracking could potentially achieve even higher effective recall by recovering momentary missed detections through trajectory prediction.

Finally, the study examined only the person detection task. Mining environments present numerous other detection challenges including vehicles, equipment, and geological hazards. Whether the observed scaling patterns generalize to these other target types remains an open question.

5.5 Future Research Directions

Several avenues merit further investigation. First, architectural modifications specifically targeting ultra-tiny detection could potentially improve upon the results achieved here. Approaches such as adding P2 detection heads operating at stride 4, incorporating super-resolution preprocessing, or applying the Normalized Wasserstein Distance loss function might yield additional gains.

Second, the inverse scaling phenomenon warrants systematic study across varying object scales. Identifying the precise scale threshold at which larger models begin to underperform would provide valuable guidance for practitioners selecting architectures for specific applications.

Third, thermal imaging fusion presents a promising direction for mining environments. Dust occlusion significantly degrades visible-light detection in active mining areas. Thermal cameras penetrate particulate interference effectively and could complement RGB detection, particularly during high-dust operations.

Fourth, on-device validation and optimization using actual Jetson hardware would

strengthen the deployment recommendations. TensorRT compilation, INT8 calibration with domain-specific data, and thermal stress testing would provide definitive performance characterization for production systems.

6 Conclusion

This study presented a systematic comparative evaluation of contemporary YOLO architectures for ultra-tiny person detection in open-pit mining environments. Through experiments spanning nine model configurations across YOLO26, YOLOv12, and YOLOv11, we examined detection performance and edge deployment feasibility for targets averaging merely 9×9 pixels substantially below conventional small object thresholds.

The findings challenge conventional assumptions about model scaling in object detection. Contrary to expectations, nano variants consistently outperformed larger counterparts across all three architectures, with YOLOv11-Nano achieving the highest overall performance: 0.973 F1-score, 0.991 recall, and 0.977 AP@50. This inverse scaling behavior appears attributable to feature representation collapse in deeper networks, where aggressive downsampling eliminates spatial information essential for ultra-tiny target localization before meaningful encoding can occur.

Among the evaluated architectures, YOLOv11 demonstrated clear superiority across both accuracy metrics and computational efficiency. YOLO26’s small-target-aware label assignment provided marginal localization improvements reflected in its leading AP@50-95 score, but did not translate to superior overall detection. YOLOv12’s attention mechanisms, despite theoretical promise for preserving spatial information, imposed computational overhead without measurable accuracy benefits for this extreme detection scenario.

From a deployment perspective, YOLOv11-Nano achieves 31 frames per second on NVIDIA Jetson Nano with FP16 precision, comfortably exceeding real-time requirements while leaving processing headroom for tracking and alert systems. Its combination of highest accuracy and fastest inference eliminates typical accuracy-efficiency trade-offs, making it the recommended configuration for mining vehicle integration.

The practical significance extends beyond technical metrics. A recall exceeding 99% means that collision avoidance systems built on this foundation would detect nearly all personnel within monitored areas. While no detection system achieves perfection, this performance level represents a meaningful safety improvement for an industry where vehicle-person collisions remain a persistent cause of fatalities.

Several limitations warrant acknowledgment: the proprietary dataset precludes independent replication, edge performance figures are estimates pending on-device validation, and the evaluation excludes temporal tracking that would further improve practical reliability. Future work should address these gaps while exploring architectural modifications

specifically targeting ultra-tiny scales, thermal-visible fusion for dust penetration, and comprehensive on-device optimization.

This work contributes empirical evidence that challenges prevailing assumptions about model scaling, demonstrates the feasibility of real-time ultra-tiny person detection on edge hardware, and provides practical guidance for practitioners developing safety systems in mining and similar industrial environments where personnel detection at extreme distances is essential.

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